

MIT OCW Thinking Notes

Unit 1 • Lecture 2

Reconstructing ideas from [MIT OCW “Principles of Chemical Science”](#) (Unit 1, Lecture 2) through intuition, confusion, and understanding.

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Introduction

These notes are based on [MIT OpenCourseWare: Principles of Chemical Science](#).

This is not a transcript. It is a reconstruction of ideas from Unit 1, Lecture 2, written through intuition, confusion, and understanding.

My Own Explanation

J.J. Thomson was studying cathode rays. Back then, people did not yet know that cathode rays were made of electrons, so he wanted to figure out what they actually were.

He did this experiment:

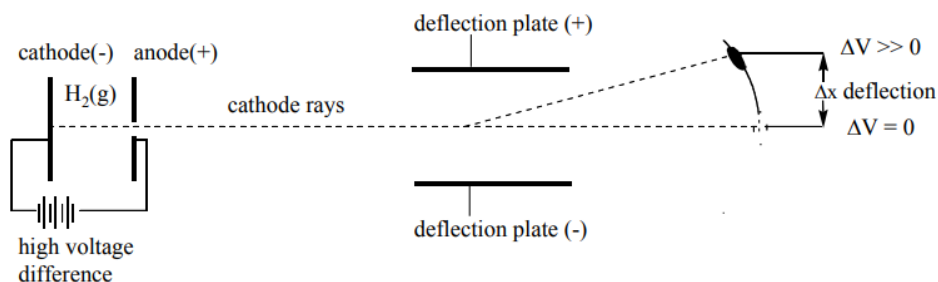


Image source: MIT OCW

At first, the plates were not charged, so the cathode ray just moved straight.

Then he charged the plates, and the ray got deflected.

This deflection is important: it means the particle does not just move forward — it gains a sideways acceleration due to an external force.

This happens because the particles feel an electric force:

$$F = qE$$

And since

$$F = ma$$

we get:

$$ma = qE$$

so:

$$a = \frac{qE}{m}$$

Now the key idea: this acceleration is perpendicular to the original motion, so it produces a sideways displacement (deflection), not just speed change.

Using the motion formula:

$$x = \frac{at^2}{2}$$

we get:

$$x = \frac{qEt^2}{2m}$$

So this deflection depends on charge-to-mass ratio, which was the key insight showing cathode rays are charged particles (**electrons**).

Rutherford Experiment

Later, Rutherford studied the atom using alpha particles.

(Alpha particles are positively charged particles. They have 2 protons and 2 neutrons.)

He fired alpha particles at a very thin gold foil.

The Experiment:

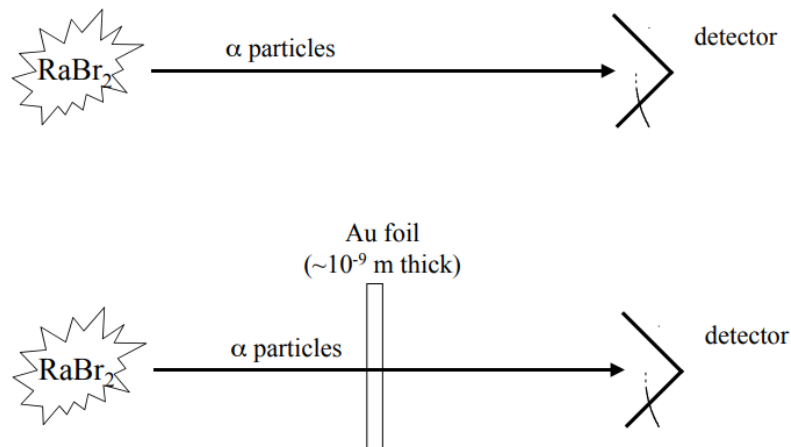


Image source: MIT OCW

In the first setup, the detector counted about 132,000 particles per minute.

Then he placed the gold foil in the middle and used detectors to see what happened to the particles.

Most of the alpha particles passed straight through.

Some were deflected.

In this experiment, roughly ~ 20 particles per minute were detected as backscattered.

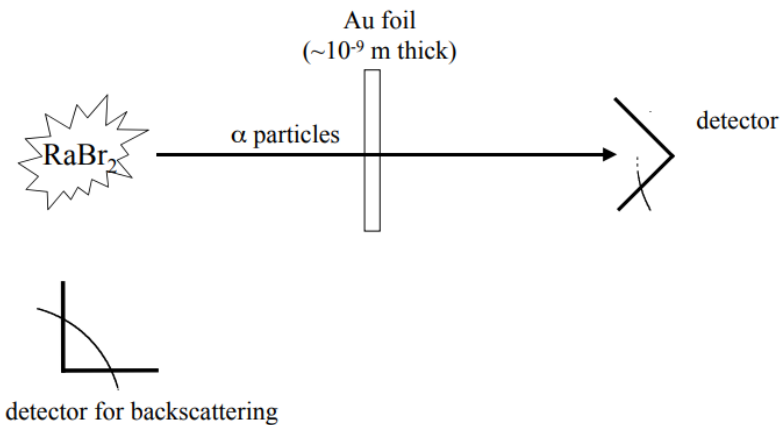


Image source: MIT OCW

That was the surprising part.

It showed that the atom is mostly empty space, and that the positive charge is concentrated in a very small part of the atom, which we now call the **nucleus**.

Probability Idea (Rutherford Scattering)

Now the key idea is how detector counts relate to probability.

If P is the probability of a given outcome (for example, backscattering), then:

$$P = \frac{\text{number of backscattered particles}}{\text{total number of particles fired}}$$

Here, "total number of particles fired" means all alpha particles sent toward the foil in the experiment. So if N_{back} is the number of backscattered particles and N_{total} is the total number of particles fired:

$$P = \frac{N_{\text{back}}}{N_{\text{total}}}$$

Now we model this geometrically.

Each gold atom contains one nucleus with effective cross-sectional area:

$$\pi r^2$$

If the foil contains N_{atoms} atoms spread over a total area A_{foil} , then the total "target area" available for scattering is:

$$N_{\text{atoms}} \cdot \pi r^2$$

So the probability becomes the fraction of target area over total foil area:

$$P = \frac{N_{\text{atoms}} \cdot \pi r^2}{A_{\text{foil}}}$$

Solving for r :

$$r = \sqrt{\frac{P A_{\text{foil}}}{N_{\text{atoms}} \pi}}$$

Since $P = \frac{N_{\text{back}}}{N_{\text{total}}}$, we can also write:

$$r = \sqrt{\frac{N_{\text{back}}}{N_{\text{total}}} \cdot \frac{A_{\text{foil}}}{N_{\text{atoms}} \pi}}$$

Summary

Thomson showed that cathode rays are negatively charged particles.

Rutherford showed that atoms have a tiny dense nucleus and are mostly empty space.

The detector counts mattered because they showed how many particles went straight, how many were deflected, and how many bounced back.

Questions i faced & confusions

Heat — and why the lamp doesn't light up in this circuit:

battery → wire → metal plate → air → metal plate → wire → lamp → wire → battery

The lamp doesn't light up because air is an insulator. However, when we heat the air, the lamp can light up. Heating increases the kinetic energy of the air particles, so they move faster and collide more strongly with each other. Some of those collisions can free electrons from atoms or molecules.

This is the mechanism:

I was thinking first that when two atoms get close enough, one atom's nucleus could attract the other atom's electron so strongly that the electron is freed, and because the atoms are moving fast, that electron is not grabbed but just freed.

But this was wrong. It turns out that energy is transferred to those electrons. What does that look like? I got a clue that maybe something is doing work there. I thought it was the nucleus of another atom doing the work on that electron. But it turns out to be way more interesting.

Another question popped up: first of all, can electrons' trajectories cross overall? And the reality is that electrons do not follow clean trajectories in the quantum world. They are described by probability clouds, meaning we can only predict the region where an electron is likely to be right now.

But can those clouds overlap? Yes.

So what does that energy transfer actually look like? The electron feels the total electric force from nearby charges. Other electrons repel it, while nuclei attract it. These forces can do work on the electron, transferring energy to it. If the transferred energy becomes larger than the electron's binding energy, the electron escapes.

But hot air alone is not enough

Why? Mechanism:

Imagine atoms colliding and creating positive ions and electrons. But if there is no pushing force, that is, no electric field, to accelerate and separate them, they would quickly combine again. So you would just have atoms, ions, and electrons floating around for a moment, instead of a sustained current.

We also need an electric field between the plates to drive the charges in opposite directions. Without the electric field, freed electrons and ions would just move randomly and recombine. So the air becomes a mixture of positive ions, free electrons, and neutral air molecules still moving around fast. Note: ionization can happen with or without breaking the molecular bond.

Positive ions move toward the negative plate. They do not travel through the wire — instead, when they reach the metal plate, they can pick up a free electron from the metallic lattice, which has many free electrons available, and become neutral again. Then new ions can be created again, and the whole process continues. Meanwhile, electrons move toward the positive plate, travel through the wire, light up the bulb, and return to the battery. The battery internally moves them from the positive terminal to the negative terminal. They then travel through the wire again, enter the positive plate, and become part of its free electrons.

This whole process happens continuously and fast, so there is no risk of the negative plate “running out” of electrons.

What determines how much air gets ionized:

- Temperature — higher temperature means higher particle energy and stronger collisions
- Electric field strength — a stronger field helps pull charges apart and can trigger more ionization
- Type of gas — some gases ionize more easily than others because different atoms hold onto their electrons with different strength. To remove an electron, enough energy must be transferred to overcome its binding energy. The required energy is called ionization energy.
- Pressure / density — because particle spacing affects collisions

All of this raises a deeper question: what actually is heat?

Heat is energy transfer.

First of all, what is temperature? It is the average kinetic energy of the particles of an object. The object consists of atoms or molecules, and those particles are always moving or vibrating. When we touch a material, its particles transfer energy to our hand's particles through interactions at the surface, and that energy transfer is what we feel as heat.

What about rigid objects? Yes, they have structure, packed tighter than liquids. But their particles still vibrate. That vibration is one of the ways thermal energy is stored and transferred.

Consider rigid object A touching rigid object B. Object A's particles transfer energy to B's particles at the contact zone, losing some energy themselves. Then, like a chain reaction, that energy spreads through object B by interactions between neighboring particles, and that is why the whole object B becomes warmer.

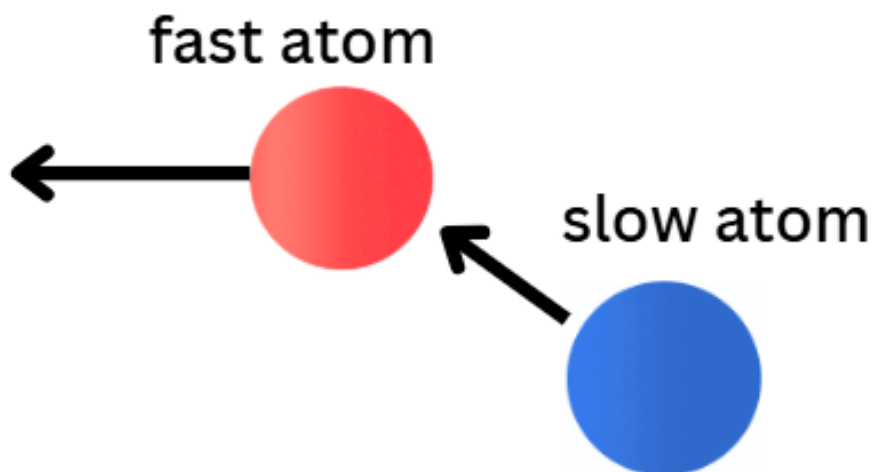
More detailed:

As atoms in object A vibrate, the electromagnetic forces they exert on nearby atoms in object B change as well. These forces can do work on object B's atoms, causing them to vibrate more strongly and gain energy too.

Imagine a cup with a hot liquid inside. Why do we feel warmth when we touch the cup? The liquid's particles transfer energy to the cup walls → heat transfers to the cup → the whole cup becomes warm → we touch the warm wall → the cup's particles transfer energy to our hand's particles.

Why does heat always flow from hot to cold, and never the other way on its own?

My first intuition was: because the hot object has more energy. When a slow-moving (cold) atom hits a fast-moving (hot) one, the fast one transfers energy to the slow one. But then I visualized a counterexample: imagine the fast-moving atom is in front, moving in a straight line, and the slow one approaches from behind on a curved path. They collide — and the fast one can actually become faster if it gets pushed from behind, while the slow one can become slower if it hits an obstacle. So “the fast one always gives energy to the slow one” is not a general rule.



The real answer is the 2nd Law of Thermodynamics: entropy cannot decrease spontaneously. What this means is that entropy describes how spread out energy is in a system, and “spontaneously” means it happens on its own, without external influence. This means that heat and energy do not spontaneously concentrate. A hot cup of coffee spreads its heat into the room — the room's energy does not spontaneously gather back to reheat the coffee.